Abdelrahman Khalifa

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Summary

Systems software engineer working in modern C++, Rust and C on Linux. Studying database internals through Carnegie Mellon's database systems course and distributed systems through the Raft paper and *Designing Data-Intensive Applications*. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), with 11 PRs merged upstream. Codeforces Expert and 4-time ECPC finalist.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI: printer status and supply levels, print queues, live updates and an Add Printer flow (cosmic-settings #2182, in review).
- Designed it as 5 crates with an async client and ran the C FFI backend as a separate varlink IPC service, so a crash in native code cannot take down the Settings app.
- Got **11 PRs merged upstream** (cups-rs, #164, #165): ported cups-rs to libcups3 with a CI matrix across architectures and CUPS versions, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Cut the desktop control app's CPU usage and latency by about 40% (measured) by waiting for ESP32 and server messages concurrently with asyncio, off the UI thread.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553); Rust Clippy: fixed `ref_as_ptr` false positives (PR #16214).

Projects

Database Systems, CMU 15-445 | *C++17* GitHub

- Thread-safe Count-Min Sketch for the BusTub database: a depth × width counter matrix with one `std::mutex` per row so inserts on different rows run in parallel; count, merge, top-k and move semantics.

Kernel Monitor | *C, Linux, ARM, QEMU* GitHub

- Character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time; a pthread user-space reader polls it.

Technical Skills

Languages: Modern C++ (C++17, STL), Rust, C, Python, SQL, Java, Bash
Computer science: Data structures and algorithms, concurrency and multithreading (mutexes, RCU, async), operating systems, memory management, networking (TCP/UDP, IPC)
Databases & distributed: Database internals (storage and page layout, buffer pool, row/column storage and compression, hash tables), SQL, DuckDB, MongoDB; Raft, replication and consistency models

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors. Treasurer, IEEE Student Branch (budget of more than $10k).
- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024.